Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	30 October 2026
Team ID	
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution, a ServiceNow scoped low-code application built in App Engine Studio on a Personal Developer Instance.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**Network Service Request Submission**	• Open the Network Request catalog item through the Service Portal (/sp) • Capture Requested For, Mobile Number, Type of Connection (New / Existing), Total Amount, Mode of Payment (UPI / Card) and Address • Provide default values: Type of Connection = New, Mode of Payment = UPI
FR-2	**Requester Information Handling**	• Select a user in the 'Opened on behalf of' field of the Requester Information variable set • Update User Name and Email when the selected user changes • Populate Phone Number when the selected user's profile contains a mobile phone number
FR-3	**Form Validation**	• Show the Existing ID field and make it mandatory only when Type of Connection is Existing (Catalog UI Policy) • Hide Existing ID when Type of Connection is New • Require Requested For, Mobile Number, Type of Connection, Total Amount, Mode of Payment and Address before submission
FR-4	**Request Record Creation**	• Start the Network Request flow when the catalog item is requested • Get the six catalog variables and create a record in the Database Tables table • Store the requested item number as the Database Number and set Approval Status to Requested
FR-5	**Request Acknowledgement**	• Send an email with subject 'Request has been Created' • Include the request number and the expected resolution time of two business working days
FR-6	**Approval Management**	• Ask for approval on the created record (reason: Waiting for approval; approver in testing: System Administrator) • Show approval records through a related list on the Database Tables record • Approved: set Assignment to = ITIL User and Approval Status = Approved • Otherwise: set Assignment to = Abraham Lincoln and Approval Status = Rejected (configured; not freshly tested)
FR-7	**Task Tracking**	• Maintain a Network Task Table with Number, Database Number, Request Number, Requested For, Approval Status, Work Status, Assigned to, Description and Work Notes • Link each task to its Database Tables record and catalog request • Current limitation: task records are created manually; the Network Request flow does not create them

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-8	**Resolution Notification**	• Run the Network Task Completion Email flow when a task's Work Status changes to Completed • Send an email with subject 'Request has been Resolved' to the Requested For user, addressed by name

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The request form is a single Service Portal catalog item with seven main variables. Default choices, user-detail auto-population and a conditional Existing ID field reduce typing and prevent incomplete submissions. Plain-language emails give the request number and expected timeframe.
NFR-2	**Security**	Uses the ServiceNow application scope, application roles, Access Control Lists (generated/default ACLs for Create, Read, Write and Delete using Allow If) and platform-managed authentication. This is a development-level configuration; ACLs must be reviewed for least-privilege access before production use. No custom encryption, firewall or security certification is claimed.
NFR-3	**Reliability**	Request processing is run by activated Flow Designer flows on a platform-managed database, and approval outcomes are recorded on the request record. Earlier runs showed missing catalog-variable warnings; a later test stored all values correctly, so wider regression testing is still recommended. An Outbox state of Ready shows an email was queued, not that it reached an inbox.
NFR-4	**Performance**	Processing is server-side in Flow Designer, and the completion flow triggers only when Work Status changes to Completed. No response-time, throughput or load-test figures have been measured.
NFR-5	**Availability**	Availability depends on the ServiceNow-hosted Personal Developer Instance. The project configures no load balancing or failover, and no production availability or SLA is claimed.
NFR-6	**Scalability**	The scoped application separates the catalog, flows and custom tables, so new catalog items, flows and tables can be added later. Scalability has not been tested with realistic request volumes and is bounded by instance and platform limits.